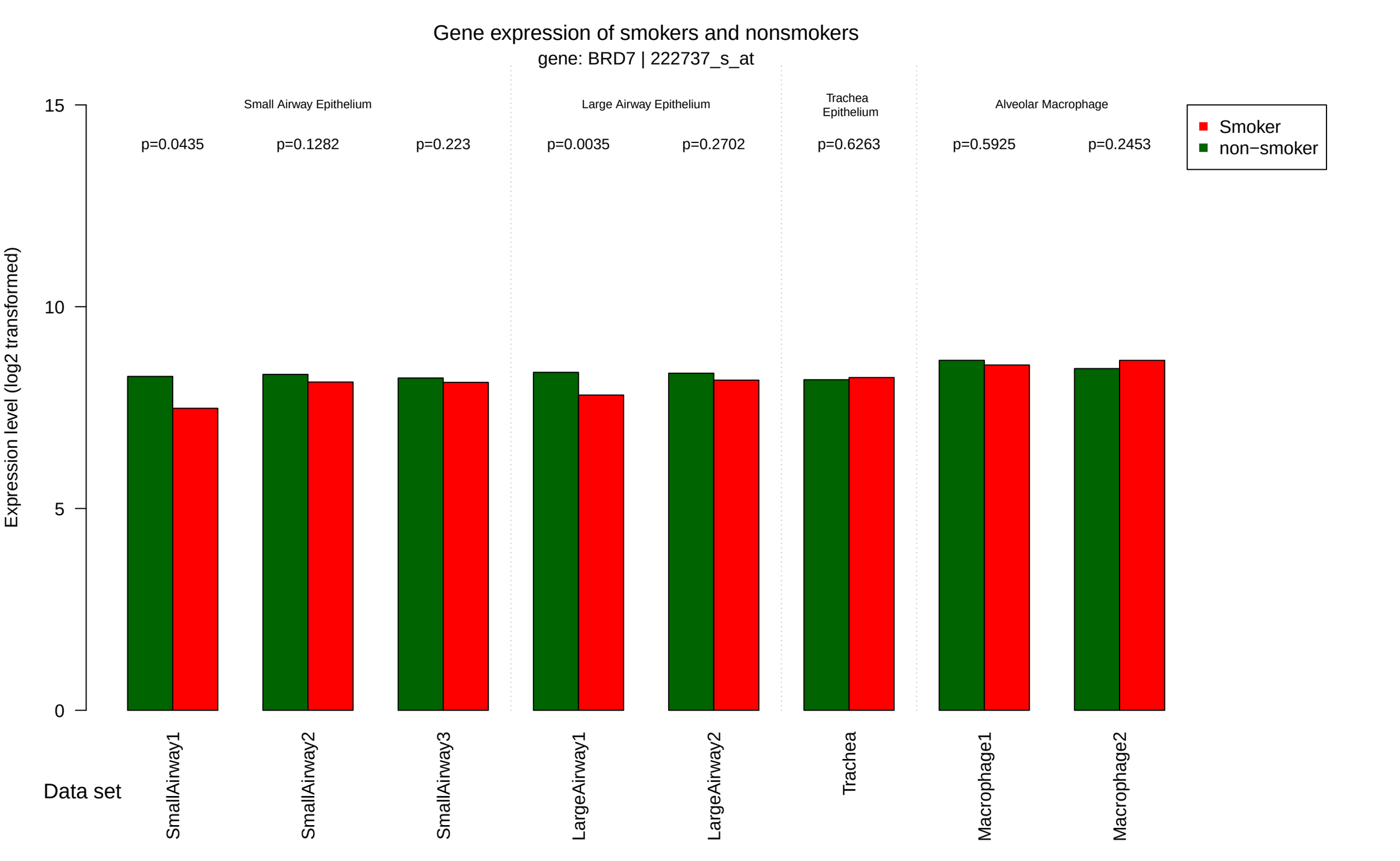
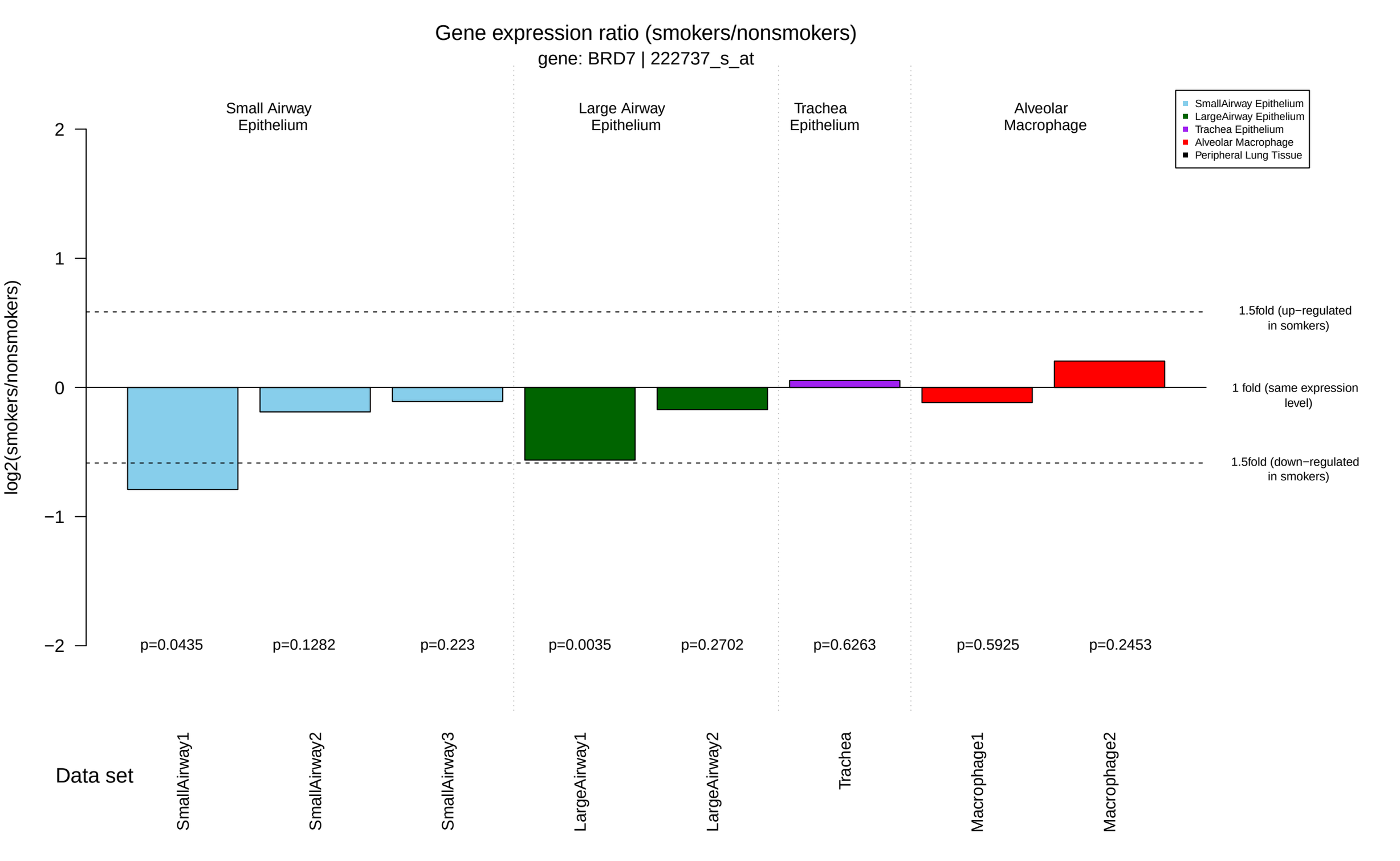
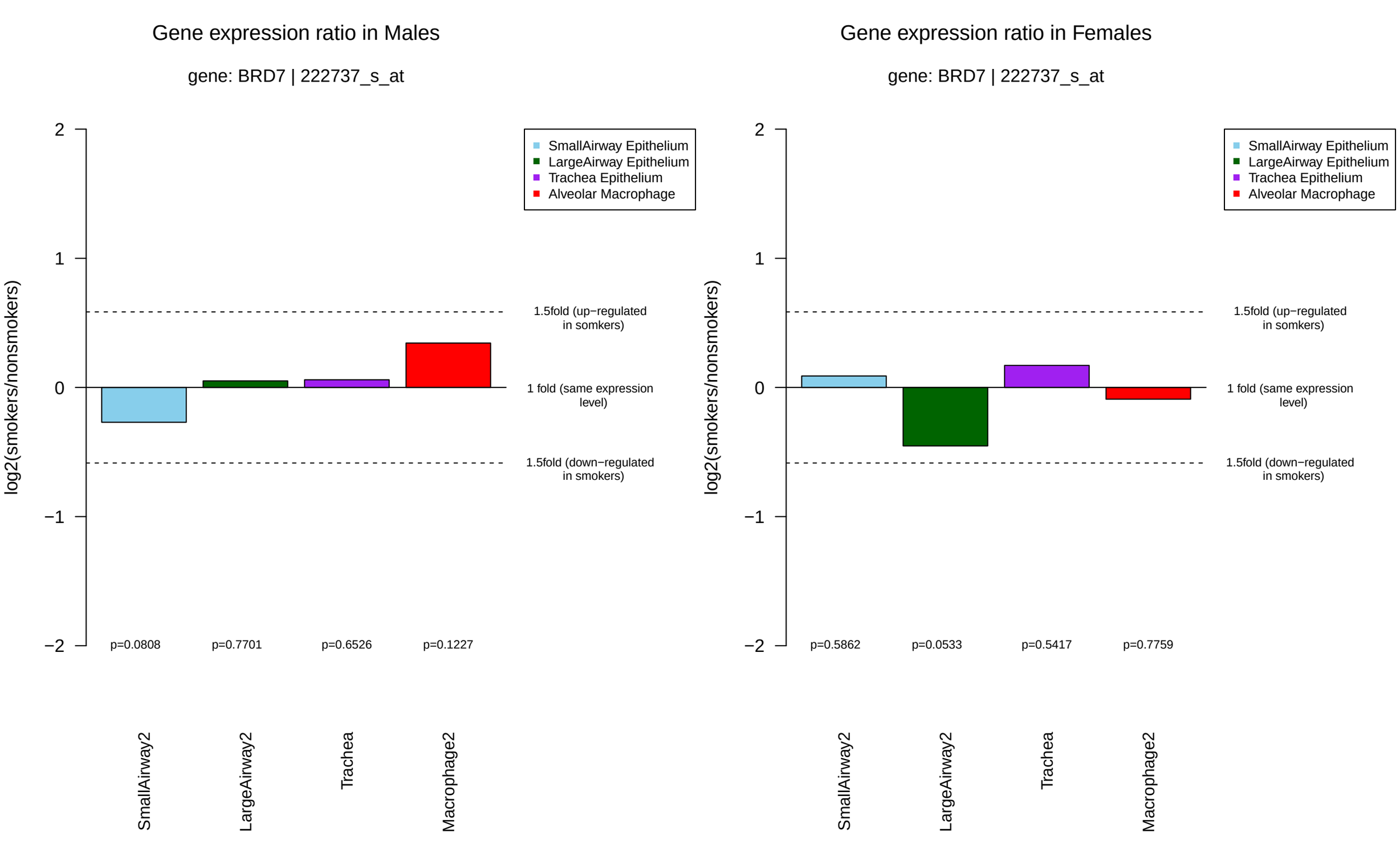
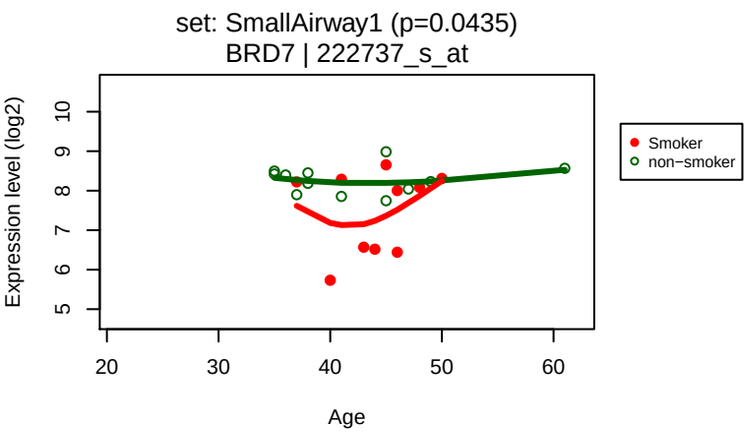
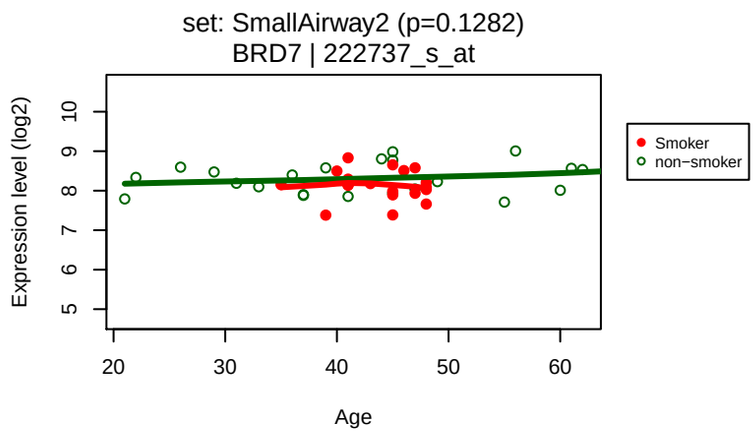
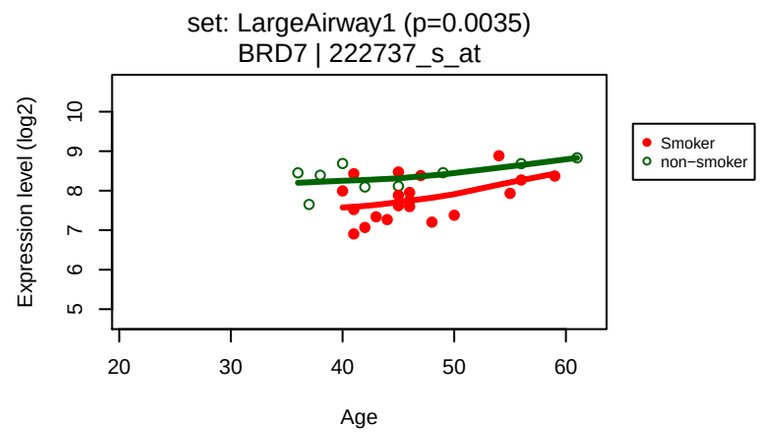

Gene expression of smokers and nonsmokers
gene: BRD7 | 222737_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0435
p=0.1282
p=0.223
p=0.0035
p=0.2702
p=0.6263
p=0.5925
p=0.2453
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

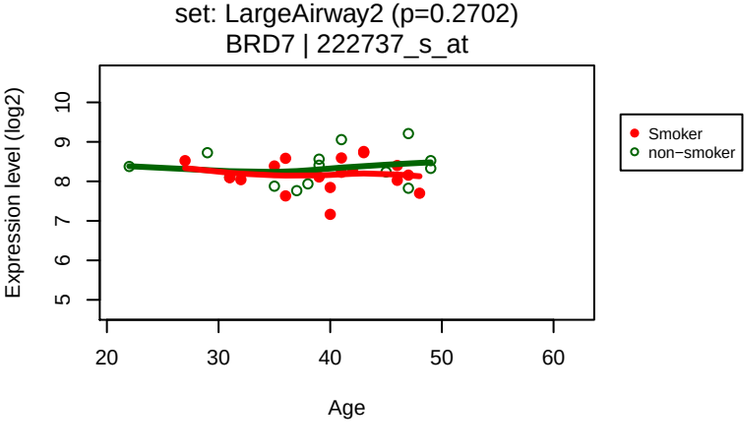

Gene expression ratio (smokers/nonsmokers)
gene: BRD7 | 222737_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0435
p=0.1282
p=0.223
p=0.0035
p=0.2702
p=0.6263
p=0.5925
p=0.2453
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

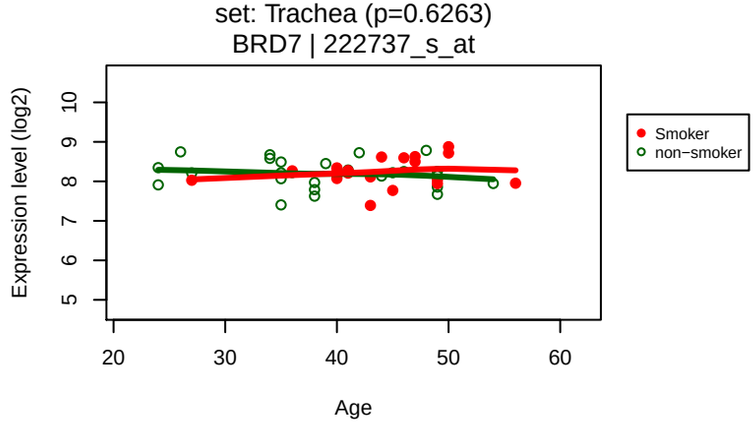

Gene expression ratio in Males
gene: BRD7 | 222737_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0808
p=0.7701
p=0.6526
p=0.1227
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: BRD7 | 222737_s_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.5862
p=0.0533
p=0.5417
p=0.7759
SmallAirway2
LargeAirway2
Trachea
Macrophage2

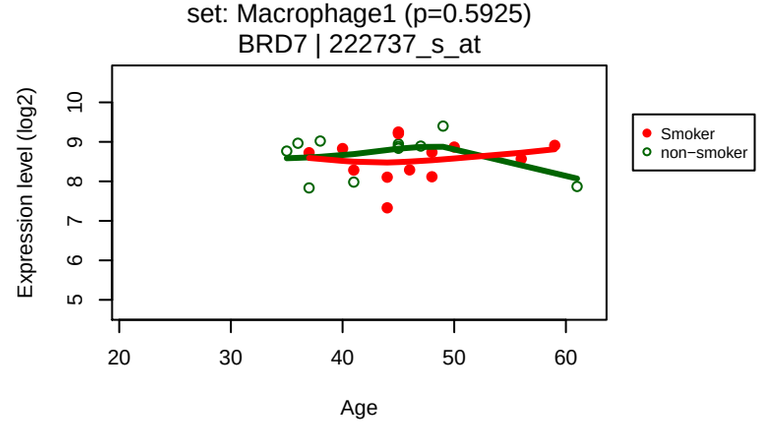

set: SmallAirway1 (p=0.0435)
BRD7 | 222737_s_at
Expression level (log2)
Age
Smoker
non−smoker

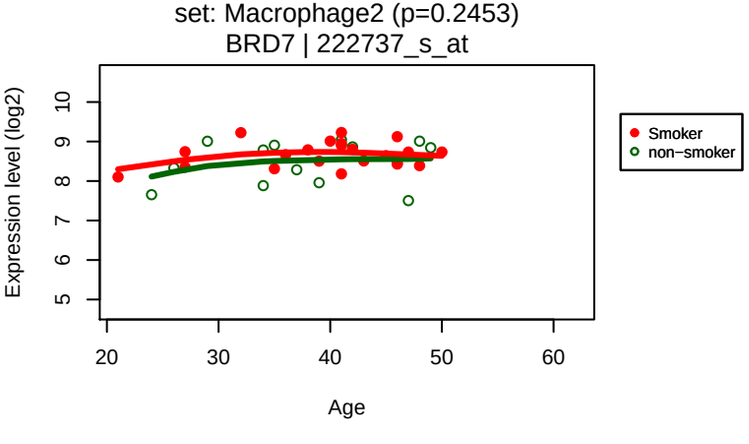

set: SmallAirway2 (p=0.1282)
BRD7 | 222737_s_at
Expression level (log2)
Age
Smoker
non−smoker

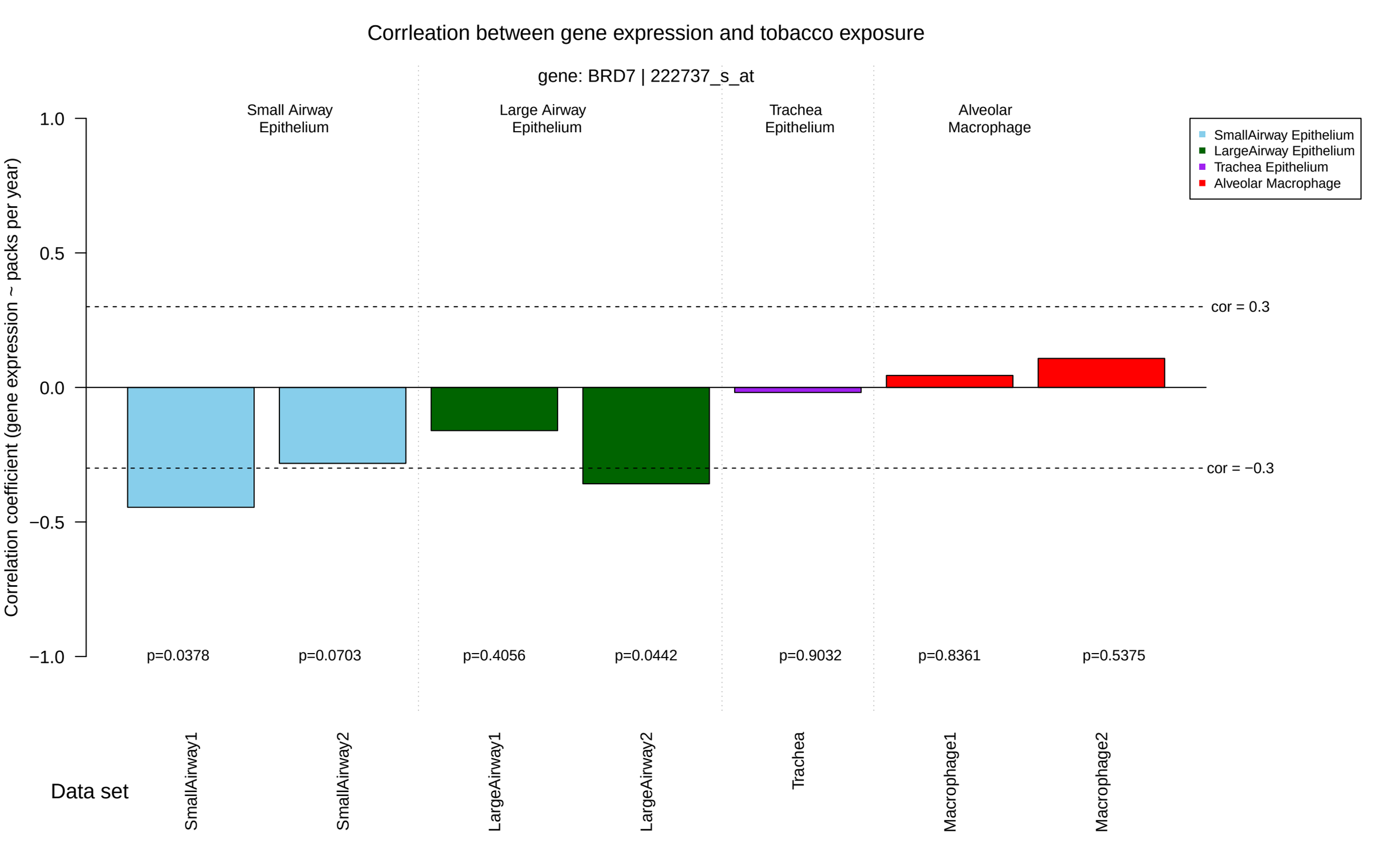

set: LargeAirway1 (p=0.0035)
BRD7 | 222737_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.2702)
BRD7 | 222737_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.6263)
BRD7 | 222737_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.5925)
BRD7 | 222737_s_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.2453)
BRD7 | 222737_s_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: BRD7 | 222737_s_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.0378
p=0.0703
p=0.4056
p=0.0442
p=0.9032
p=0.8361
p=0.5375
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2